# Ruiyang DAI

Innovative Numerical Methods • High-Performance Computing
Applied Mathematics

## Contact

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Pau, France

## Summary

I am a mathematical engineer specialized in Innovative Numerical Methods and High-Performance Computing for partial differential equations.

I am passionate about exploring new ideas and integrating modern technologies with classical numerical techniques to develop efficient and reliable numerical methods.

With strong organizational skills, I can manage multiple projects simultaneously, assess different methodologies, and evaluate the strengths and limitations of each approach to identify the most effective solutions.

## Expertise

- Numerical methods for partial differential equations
- Scientific Machine Learning

## Programming Languages

Fortran, C, C++, Python

## Languages

- Chinese (Native)
- English (Fluent)
- French (Fluent)
- Italian (Basic)

## Professional Experience

**Starting research position** 12/2025 – Present
*INRIA Bordeaux*

- Acceleration of full-waveform inversion for seismic applications.
- Development of numerical methods for wave propagation.

**Postdoctoral Fellow** 01/2025 – 09/2025
*INRIA Paris & Sorbonne University*

- Machine learning and evolution equations.
- Machine learning for the Fokker-Planck-Landau equation.

**Postdoctoral Fellow** 09/2021 – 12/2024
*Sorbonne University*

- Moment methods for Vlasov equations with a non-constant magnetic field.
- Spectral methods for the Vlasov–Poisson system.

**PhD in the ERC HEXTREME project** 10/2016 – 08/2021
*UCLouvain*

- Generalized preconditioners for domain decomposition methods applied to Helmholtz problems.
- Numerical methods for wave propagation.

## Teaching Experience

**Sorbonne University** 2024 – 2025
*Level: Master's degree*
Subject areas: Numerical analysis, Linear algebra.
Course name: Foundations of numerical methods.
Instructor: Albert Cohen.

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Subject areas: Numerical analysis, Linear algebra.
Course name: Foundations of numerical methods.
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**Université Paris Cité** 2023 – 2024
*Level: Master's degree*
Subject areas: Partial differential equations.
Course name: Introduction to partial differential equations.
Instructor: Luca Alasio.

## Education

**PhD** 2016 – 2021
*Engineering Sciences*
Université catholique de Louvain, Louvain-la-Neuve, Belgium.

**Master's degree** 2014 – 2016
*Applied Mechanics*
École Centrale de Nantes, Nantes, France.

**Bachelor's degree** 2010 – 2014
*Mechanics*
Wuhan University, Wuhan, China.